

Muhammad Abdullah Azzam

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ABOUT ME

Ph.D. student and water resources professional with expertise in computational hydrology, hydrologic and hydraulic studies, coupled surface-groundwater modeling, high-performance geospatial processing, and open-source hydrologic software development. Interested in developing reproducible modeling tools and open-source solutions for drought, flood, runoff, groundwater, and water-resources management applications.

EDUCATION

Doctorate in Civil Engineering Jan 2024 – Present

New Mexico State University

Research: Coupled hydrologic modeling, distributed computation, and regional drought and groundwater assessment. Expected completion: 2027.

Master of Civil Engineering 2020 – 2023

NED University of Engineering & Technology

Thesis: Assessment & improvement of QNSPECT for sediment and NPS pollutant load estimation.

Bachelor of Urban & Infrastructure Engineering 2016 – 2019

NED University of Engineering & Technology

EXPERIENCE

Graduate Intern May 2026 – Aug 2026

Lawrence Livermore National Laboratory, Livermore, CA

- Evaluated and optimized HPC data pipelines for scientific deep learning workflows, benchmarking storage formats, I/O performance, compression, and distributed training across CPU and GPU/APU systems.

Research Assistant Jan 2024 – Present

HydroCS Lab, New Mexico State University, Las Cruces, NM

- Building scalable scientific software for hydrologic modeling, including MPI-parallel VIC-MODFLOW 6 coupling, ParFlow workflows, and high-performance raster and geospatial processing tools.
- Integrating coupled surface-water-groundwater models, Earth-observation data, and large geospatial datasets for drought, groundwater-change, and regional/transboundary water-resources assessment.

Water Resources Engineer Aug 2021 – Dec 2023

Techno-Consult International, Karachi, PK

- Conducted hydrologic, hydraulic, flood-risk, drainage, dam, sediment-transport, water-supply, and remote-sensing studies for water-resources and infrastructure projects.

Research Assistant Oct 2019 – Feb 2021

NED University of Engineering & Technology, Karachi, PK

- Supported GIS, remote-sensing, hydrologic modeling, water-quality, sediment, and nonpoint-source pollution studies.

RESEARCH STUDIES

New Mexico Statewide Drought Vulnerability Analysis

Developed a physically based drought vulnerability framework for New Mexico using VIC-MF6, observed and projected climate data, distributed computing, and a statewide Drought Vulnerability Index. The published report evaluates historical and future regional vulnerability. [report](#)

VIC-MF6 Coupled Surface-Groundwater Modeling Framework

Developing an MPI-parallel, two-way coupling of VIC and MODFLOW 6 for land-surface-groundwater simulation. The framework supports reproducible regional assessment of water availability, drought, and groundwater feedbacks, with numerical verification and scalable workflows. [repo](#) | [report](#)

GCN10: Global 10 m Curve Number Dataset and Processing Framework

Developed a global Curve Number data product and C/MPI processing framework for large-scale raster production, runoff modeling, and watershed analysis. Released open data and software with catalog access. [paper](#) | [repo](#) | [dataset](#) | [catalog](#) | [zenodo](#)

Open-Source Hydrologic Software for GRASS GIS and QGIS

Developed and contributed open-source tools for Curve Number generation, runoff estimation, time of concen-

tration, and Manning's roughness raster generation. `r.curvenumber` | `r.runoff` | `r.timeofconcentration` | `manning-qgis`

USGS Mesilla Aquifer and Transboundary Aquifer Assessment Program

Developing a ParFlow-CLM model of the Mesilla and transboundary aquifer system to assess groundwater availability and surface-subsurface interactions. Building reproducible climate, terrain, geology, soil, and well-data workflows. `repo`

TORRENT Flood Modeling on HPC Systems

Applying TORRENT Flood Model on HPC systems to support large-domain inundation modeling and flood-hazard assessment. Working on model and workflow improvements for scalable simulations.

NSF Dynamics of Integrated Socio-Environmental Systems (DISES)

Collaborating on water availability assessment for agricultural watersheds using SWAT+, focusing on climate variability, land-use change, hydrologic response, and watershed-scale water-resources planning.

Curve Number Generator QGIS Plugin

Co-developed the global algorithm for the **QGIS Curve Number Generator Plugin**, an open-source tool for hydrologic runoff modeling. The plugin source is available through the public `repo`.

Fourier Transform in Hydrology

Explored the potential of **Fast Fourier Transform on hydrologic time series** to extract meaningful hydrologic signals and reconstruct streamflow components.

Comparison of SWAT and SWAT+

Analyzed performance differences between SWAT and SWAT+ models in a highly managed agricultural watershed. Findings were presented at the ASABE Annual International Meeting 2023.

Critical Analysis of QNSPECT by NOAA

Master's thesis assessing QNSPECT, an enhanced GIS-based methodology for sediment and pollutant load estimation in watersheds.

Water Quality through Remote Sensing

Authored "Estimation of Suspended Sediment Concentration of Keenjhar Lake through Remote Sensing" published in Engineering Proceedings, 2022.

GIS-Based Water Management

1. Developed time-series-based crop water requirement estimates using Sentinel imagery for arid regions.
2. Created GIS-based groundwater potential zones maps for multiple cities using the Analytic Hierarchy Process.
3. Conducted 1D/2D hydraulic modeling of the Machar and Hamal Lake plains, validated by Google Earth Engine flood-inundation maps for historical and projected climate-change events.

Upper Rio Grande Groundwater and Drought Assessment

Study underway to assess drought impacts on groundwater storage and availability in the Upper Rio Grande Basin through two-way land-surface-groundwater coupling.

PUBLICATIONS & REPORTS

Peer-Reviewed Publications

1. Azzam, M. A., & Cho, H. *GCN10: An MPI-parallelized framework for processing global curve number rasters for hydrologic modeling*. SoftwareX, 34, 102725, 2026. DOI | `repo`
2. Azzam A., Uddin H., Mannan U. *Estimation of Suspended Sediment Concentration of Keenjhar Lake through Remote Sensing*. Engineering Proceedings, 2022, 22, 20. DOI

Conference Papers

1. Cho, H., & Azzam, M. A. *A Coupled Hydrologic Modeling Framework for Drought Evaluation in New Mexico*. Proceedings of the 36th Conference of the Japan Association for Information and Earth Science, pp. 001–002, June 2025. `paper`

Technical and Scientific Reports

1. Cho, H., & Azzam, A. *Development of a Computational Framework for Statewide Coupled Surface Water and Groundwater Modeling*. New Mexico Water Resources Research Institute Technical Completion Report No. 412, New Mexico State University, May 2026. `report`
2. Cho, H., & Azzam, A. *New Mexico Statewide Drought Vulnerability Analysis Under Future Climate Change Scenarios Using a Physically Based Coupled Model*. New Mexico Water Resources Research Institute Technical Completion Report No. 413, New Mexico State University, June 2026. `report`

Manuscripts in Preparation

1. Azzam, M. A., & Cho, H. *GCN10: A New Global High-Resolution Gridded Curve Number for Improved Hydrological Studies*. Manuscript in preparation. [dataset](#) | [repo](#)
2. Azzam, M. A., & Cho, H. *An MPI-Parallel Two-Way Coupling of VIC and MODFLOW 6 for Land-Surface–Groundwater Simulation*. Manuscript in preparation. [repo](#)
3. Azzam, M. A., & Cho, H. *New Mexico Statewide Drought Vulnerability Analysis Under Future Climate Scenarios Using a Physically Based Coupled Model*. Manuscript in preparation.

Study Underway

1. Azzam, M. A., & Cho, H. *Assessing Drought Impacts on Groundwater Storage and Availability in the Upper Rio Grande Basin through Two-Way Land-Surface–Groundwater Coupling*. Study underway.

DATASETS & OPEN-SOURCE SOFTWARE

1. **GCN10: Global 10 m Curve Number Dataset** – Global open NRCS/SCS Curve Number raster dataset for runoff modeling and watershed analysis. [dataset](#) | [catalog](#) | [zenodo](#)
2. **GCN10** – MPI-parallelized framework for processing global Curve Number rasters. [repo](#)
3. **gcn10py** – Python interface to the high-performance C-based GCN10 executable. [repo](#)
4. **VIC–MF6** – MPI-parallel, two-way VIC and MODFLOW 6 coupling framework with numerical verification and reproducible examples. [repo](#)
5. **TAAP ParFlow workflows** – Model-input preparation and CLM forcing for the Mesilla Aquifer. [repo](#)
6. **r.curvenumber** – GRASS GIS addon for Curve Number raster generation. [docs](#)
7. **r.runoff** – GRASS GIS addon for runoff depth, runoff volume, and peak discharge estimation. [docs](#)
8. **r.timeofconcentration** – GRASS GIS addon for time-of-concentration estimation. [docs](#)
9. **Manning’s Roughness Generator** – QGIS plugin for generating 2D Manning’s roughness coefficient rasters from land-cover data. [qgis](#) | [repo](#)
10. **Curve Number Generator QGIS Plugin** – Open-source QGIS plugin for Curve Number generation; contributed global algorithm/workflow. [qgis](#) | [repo](#)
11. **WaveFlow** – Hydrologic time-series analysis workflow using Fast Fourier Transform methods. [repo](#)

PRESENTATIONS

1. Cho, H., & Azzam, M. A. *Hydrologic Modeling, GIS, and Computational Methods for Water Resources Management*. UCOWR/NIWR Annual Water Resources Conference, San Antonio, TX, June 8, 2026.
2. Azzam, M. A., & Cho, H. *GCN10: Global 10 m Curve Number Dataset and High Performance Raster Processing Framework*. New Mexico Water Conference, Las Cruces, NM, October 22, 2025.
3. Fernald, A., Carroll, K., Cho, H., Granados, A., Azzam, M. A., & Garza, I. *Transboundary Aquifer Assessment Program*. New Mexico Water Conference, Las Cruces, NM, October 22, 2025.
4. Cho, H., & Azzam, M. A. *Special Lecture: A Coupled Hydrologic Modeling Framework for Drought Evaluation in New Mexico*. GEOINFORUM-2025, Osaka Metropolitan University, Osaka, Japan, June 27, 2025. [abstract](#)
5. Azzam, M. A., & Cho, H. *GCN10: A High-Resolution Global Curve Number Dataset and a Parallelized Framework for Large-Scale Raster Operations*. AWRA Spring Conference, Anchorage, AK, April 28, 2025.
6. Azzam, M. A., & Cho, H. *Coupling of Parallelized Surface Water and Groundwater Models for Statewide Hydrologic Modeling*. AGU Annual Meeting, Washington, D.C., December 12, 2024.
7. Shreesh, H., Azzam, M. A., & Cho, H. *Assessing Water Availability in the El Rito Watershed Using SWAT and Parameter Regionalization*. New Mexico Water Conference, Pojoaque, NM, November 5, 2024.
8. Azzam, M. A., & Cho, H. *Global High-Resolution Curve Number for Hydrological Studies*. New Mexico Water Conference, Pojoaque, NM, November 5, 2024.
9. Azzam, M. A., Cho, H., & collaborators. *Comparison of SWAT and SWAT+ in a Highly Managed Agricultural Watershed*. ASABE Annual International Meeting, 2023.

RESEARCH SUPPORT

1. **NM WRRI Student Water Research Proposal** – *Improving Runoff Estimation Through the Development of a Global High-Resolution Curve Number Dataset*. Submitted September 30, 2025; student proposer: Abdullah Azzam; faculty advisor: Dr. Huidae Cho.
2. **USGS WRIA 104(b) / NM WRRI Research** – *New Mexico Statewide Drought Vulnerability Analysis Under Future Climate Change Scenarios Using a Physically Based Coupled Model*. Research assistant contributing to VIC–MF6 model development, climate-data processing, drought vulnerability analysis, and coupled surface–groundwater simulation.
3. **Transboundary Aquifer Assessment Program** – Developing ParFlow–CLM model inputs, climate forcing, and reproducible hydrogeologic workflows for the Mesilla Aquifer.
4. **NSF DISES-Related Watershed Modeling Research** – Contributing to SWAT+ water availability assessment under climate and land-use change in agricultural watersheds.

INDUSTRY PROJECTS

Water Resources Engineering & Management

- Development of a 15-year water resources master plan of Quetta City for the World Bank by identifying potential surface-water sources, preserving and recharging groundwater, and improving water supply throughout the city.
- Development of a comprehensive working paper identifying the underlying reasons for flooding in Sindh and proposing an engineering & management working plan for the Pakistan Engineering Council.
- Comprehensive flood-risk assessment under climate-change scenarios and impact assessment in the Indus River Basin, especially Punjab, Pakistan, for CARE for South Asia by the World Bank and ADPC in joint venture with Royal HaskoningDHV, Netherlands.
- Developed the post-development drainage plan on behalf of the pre-development flood-risk assessment for the Silk Garden housing scheme in Surjani Town.

Computational Hydrology

- Hydrologic and hydraulic modeling for extreme events at the auxiliary spillway of Nai Gaj Dam to ensure safety against tailwater and backwater effects.
- Hydrologic study & modeling of rainfall-runoff patterns for 82 major streams and rivers to develop design flood hydrographs across the 110 km long alignment of 650 MGD water supply pipelines for Karachi Bulk Water Supply Scheme IV (K-IV). Also performed hydraulic studies and developed stream-bed erosion and sediment-transport hydraulic models to assess safe pipeline depth for multiple streams and return periods.
- Hydrologic and sediment modeling for Yellow River Watershed, Indiana, and Maumelle River, Arkansas using SWAT.
- Comprehensive water-management studies involving statistical analysis of hydrometeorological data, hydrologic & hydraulic modeling of basins & rivers, development of inundation and impact assessments, and preparation of mitigation plans including engineering solutions.

Dams & Hydraulic Structures

- Complete hydrologic and reservoir operations studies to estimate floods for several return periods as well as PMF, develop hydrographs, and develop Area-Capacity-Elevation relationships of Halak Dam, Barkhan Dam, Sorra Dam, Mangi Dam, and Bibi Nani Dam.
- Flood modeling and analysis of failure and damage of coffer dams and the main dam embankment due to the flooding in July 2022.
- Design of culverts and cross-drainage structures for 10 km long access road and 52 km long Nai Gaj Dam Project irrigation canal.

Earth Observation & Spatial Data Sciences

- Flood mapping and damage impact assessment using large-scale remotely sensed data acquisition and processing in GEE, SNAP, and QGIS after the devastating monsoon of 2022.
- Development of river meandering patterns and identification of changes in the floodplain and land cover of the Indus River from 1975 to 2022.
- GIS-based soil erosion estimation for water-conservation analysis to estimate reduction in storage capacity and reservoir operations for the lake for an outflow of 260 and 650 MGD for the K-IV Project.

SKILLS

Modeling Tools – VIC, MODFLOW 6, ParFlow-CLM, SWAT/SWAT+, HEC-HMS, HEC-RAS, TRITON, TORRENT

Programming and Computing – Python, C, R, Shell scripting, MPI, Linux, Git, Docker, HPC (MPI, Slurm, Flux Frameworks) Fortran (beginner), Julia (beginner), Tcl (beginner)

GIS and Spatial Analysis – QGIS, GRASS GIS, ArcGIS Products, SAGA, Global Mapper, Surfer, Google Earth Engine

Remote Sensing – ENVI, ERDAS Imagine, SNAP, Agisoft Metashape, Pix4D